- Production planning
  - Entities
    - Periods
      - Multi-period
        - T = {t}
    - Products
      - Single-product
      - Multi-product
        - P = {p}
    - Populate array/matrix
  - Parameters
    - Initial inventory
      - I_0
      - I_{0,p}
    - Cost
      - Unit production cost
        - labour
          - Regular time workers
            - c_t
            - c_{t,p}
          - Over time workers
            - o_t
            - o_{t,p}
        - unit material cost
          - m_t
          - m_{t,p}
      - Unit holding cost
        - h_t
        - h_{t,p}
      - Set-up cost
        - s_t
        - s_{t,p}
      - Shortage cost
        - g_t
        - g_{t,p}
    - Same for all t or different? Populate array
    - Demand
      - d_t
      - d_{t,p}
    - Labour hour required
      - h_t
      - h_{t,p}
    - Limitations
      - Labour hour limit
        - UB^L_t
        - UB^L_{t,p}
      - Material budget limit
        - UB^P_t
        - UB^M_{t,p}
    - Same for all t and p or different? Populate matrix
  - Decision
    - How many to produce
      - by regular hour labour
        - x_t
        - x_{t,p}
      - by over time labour
        - y_t
        - y_{t,p}
    - Inventory to hold
      - z_t
      - z_{t,p}
    - To produce or not
      - u_t
      - u_{t,p}
    - Shortage
      - v_t
      - v_{t,p}
  - Objective function
    - Total regular hour labour cost
      - \sum_t c_tx_t
      - \sum_t \sum_p c_{t,p} x_{t,p}
    - Total over time labour cost
      - \sum_t o_t y_t
      - \sum_t \sum_p o_{t,p} y_{t,p}
    - Total holding cost
      - \sum_t h_t z_t
      - \sum_t \sum_p h_{t,p} z_{t,p}
    - Total set-up cost
      - \sum_t s_t u_t
      - \sum_t \sum_p s_{t,p} u_{t,p}
    - Total shortage cost
      - \sum_t g_t v_t
      - \sum_t \sum_p g_{t,p} v_{t,p}
    - Total material cost
      - \sum_t m_t x_t
      - \sum_{t,p} m_{t,p} x_{t,p}
    - Sum of selected total costs
    - \min
    - Sum of selected total costs
  - Constraints
    - Explicit constraints
      - Demand of production
        - for each t: v_t + x_t + y_t = u_{t-1} - u_t (at least \leq, exactly =, at most \geq) d_t
        - for each t, for each p: v_{t,p} + x_{t,p} + y_{t,p} = u_{t-1,p} - u_{t,p} (at least \leq, exactly =, at most \geq) d_{t,p}
      - Inventory balance
        - for each t: u_t = u_{t-1} + x_t - d_t + y_t - v_t
        - for each t, for each p: u_{t,p} = u_{t-1,p} + x_{t,p} - d_{t,p} + y_{t,p} - v_{t,p}
      - Labour hour limit
        - for each t: UB^L_t \geq h_t x_t + h_t y_t
        - for each t, for each p: UB^L_{t,p} \geq h_{t,p} x_{t,p} + h_t y_{t,p}
      - Material budget limit
        - for each t: UB^M_t \geq m_t x_t + m_t y_t
        - for each t, for each p: UB^M_{t,p} \geq m_{t,p} x_{t,p} + m_{t,p} y_{t,p}
    - Implicit constraints
      - Production is possible only if set-up is "true"
        - for each t: y_t + x_t \leq D_t u_t
        - for each t, for each p: y_{t,p} + x_{t,p} \leq D_{t,p} u_{t,p}
    - \forall t \in T
    - \forall t \in T, \forall p \in P

- Forall: null → T,P; T → P; P → T; T,P → null
- Sum: T,P → null; P → T; T → P; null → T,P